Lesson 3 - Gradients

⇒ In simple 2-d dy/dx was a line, which will become a plane in 3-d and so-on.

Tangent plane

⇒ We find tangent plane in 3-d by fixing one var at a time & finding 2 tangent lines.

⇒ With 2 tangent lines, we find the tangent plan.

eg $f(x,y) = x^2 + y^2$

$$\frac{d}{dx}(f(x,4)) = 2x \qquad \frac{d}{dy}(f(4,y)) = 2y$$

Partial Derivatives

⇒ Partial derivative is a derivative with respect to 1 variable while keeping all other vars constant.

eg $f(x, y, z) = \text{---}$

$\frac{\partial}{\partial x} f(x, y, z)$

partial derivative wrt x

considered as constants

$f(x, y) = x^2 + y^2$

$\frac{\partial f}{\partial x} = 2x$ $\qquad$ $\frac{\partial f}{\partial y} = 2y$

eg $\quad f(x, y) = 3x^2 y^3$

$$\frac{\partial f}{\partial x} = 6xy^3$$

$$\frac{\partial f}{\partial y} = 9x^2y^2$$

Gradients

⇒ Gradient is the vector of all partial derivatives.

Gradient of $f(x_1, x_2 \ldots x_n)$

$$= \begin{bmatrix} \partial f/\partial x_1 \\ \partial f/\partial x_2 \\ \vdots \\ \partial f/\partial x_n \end{bmatrix} = \nabla f$$

eg find gradient of $f(x,y) = x^2 + y^2$ at $(2,3)$

$$\nabla f = \begin{bmatrix} \partial f / \partial x \\ \partial f / \partial y \end{bmatrix} = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

$$\nabla f_{(2,3)} = \begin{bmatrix} 2 \cdot 2 \\ 2 \cdot 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

Similar to how solving the derivative for single variable function gives us the minima / maxima information.

↓

Solving for $\nabla f = 0$ i.e. solving for gradient to be equal to zero vector <u>gives us the minima / maxima</u> in <u>higher dimensions</u>

⇒ eg Sauna problem [5×5 case]

$$T(x,y) = 85 - \frac{1}{90} x^2(x-6)\, y^2(y-6)$$

⇒ Now to find the coldest place in sauna →

$$\nabla f = \begin{bmatrix} \partial f/\partial x \\ \partial f/\partial y \end{bmatrix}$$

$$\frac{\partial f}{\partial x} = \frac{-1}{90} y^2(y-6)\,(3x^2 - 12x) = 0$$

$\boxed{x = 0, 4}$ $\boxed{y = 0, 6}$

$$\frac{\partial f}{\partial y} = \frac{-1}{90} x^2(x-6)\,(3y^2 - 12y) = 0$$

$\boxed{y = 0, 4}$ $\boxed{x = 0, 6}$

Now to find whether they are minimas or maximas, lets check their second derivatives.

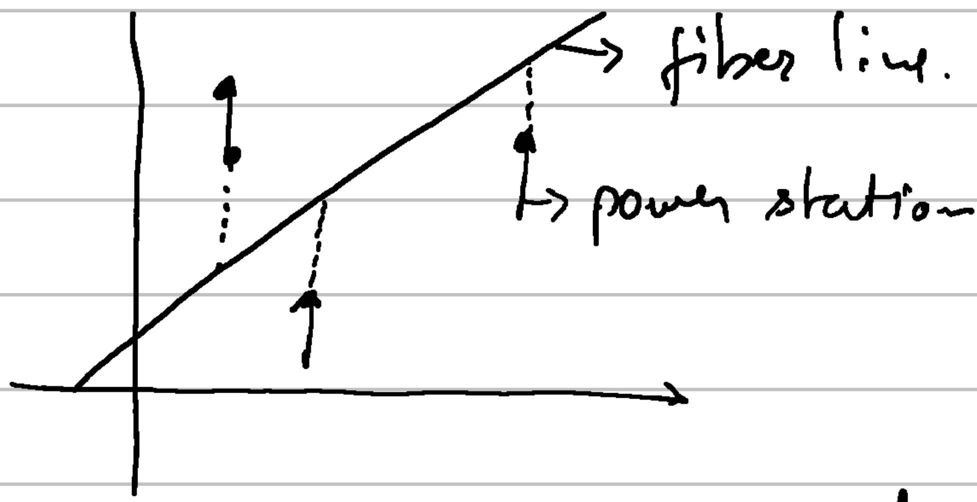
$$\frac{\partial'' f}{\partial x''} = \frac{-1}{90} y^2(y-6)(6x-12) \quad -\frac{1}{\cancel{90}_{10}} \cdot \cancel{4} \cdot (-3) \cdot 6 \quad \frac{18}{10}$$

$$\frac{\partial'' f}{\partial y''} = -\frac{1}{90} x^2(x-6)(6y-12)$$

		$\frac{\partial'' f}{\partial x''}$, $\frac{\partial'' f}{\partial y''}$	
@	$(0,0)$	$\rightarrow$ $0, 0$	[maxima]
@	$(0,4)$	$\rightarrow$ $-ve, 0$	[maxima]
@	$(4,0)$	$\rightarrow$ $0, -ve$	[maxima]
@	$(4,4)$	$\rightarrow$ $\frac{18}{10}, \frac{18}{10}$	[minima] ✓

$x=6$, $y=6$ discarded as out of bounds for the given problem.

⇒ Powerline example



Reduce the cost of putting down the fiber for power stations.

Wires can only be laid parallel to y-axis to connect fiber line & power station.

cost = square of wire length.

$$\Rightarrow \quad y = mx + b$$

$$P = (x_1, y_1), \; (x_2, y_2) \; \ldots \; (x_n, y_n)$$

$$C_1 = (y_1 - mx_1 - b)^2$$

$$\vdots$$

$$C_n = (y_n - mx_n - b)^2$$

$$\Sigma \text{Cost} = \sum_{i=1}^{n} C_q = (y_1 - mx_1 - b)^2 + \cdots$$

This cost function becomes our function to minimize.

Lets find its gradient →

$$\nabla C = \begin{bmatrix} \partial C / \partial m \\ \partial C / \partial b \end{bmatrix}$$

$$\frac{\partial C}{\partial m} = -2x_1(y_1 - mx_1 - b) - 2x_2(y_2 - mx_2 - b) \ldots$$

$$\frac{\partial C}{\partial b} = -2(y_1 - mx_1 - b) - 2(y_2 - mx_2 - b).$$

Now, solving these for particular (x_1, y_1) -- would yield us our answer. Lets see if a general form exists.

$$\frac{\partial C}{\partial m} = 0$$

$$\Rightarrow x_1 y_1 - m x_1^2 - b x_1 + x_2 y_2 - m x_2^2 - b x_2 - \cdots$$

$$x_1^2 m + x_1 b + x_2^2 m + x_2 b \cdots = x_1 y_1 + x_2 y_2 \cdots$$

$$\boxed{m(x_1^2 + x_2^2 \cdots) + b(x_1 + x_2) = x_1 y_1 + x_2 y_2 \cdots}$$

$$\Rightarrow \frac{\partial C}{\partial b} = 0$$

$$\boxed{m(x_1 + x_2 \cdots) + nb = y_1 + y_2 \cdots}$$

Yeah, the general form does not seem to be pretty. .

eg $(1,2)$ $(2,5)$ $(3,3)$

lets use the general form.

$$m(1+4+9) + b(1+2+3) = 2+10+9$$

$$14m + 6b = 21$$

$$6m + 3b = 10 \quad \times 2$$

$$14m - 12m = 1$$

$$m = 0.5$$

$$6b = 21 - 7$$

$$b = \frac{14}{6} \quad \frac{7}{3}$$